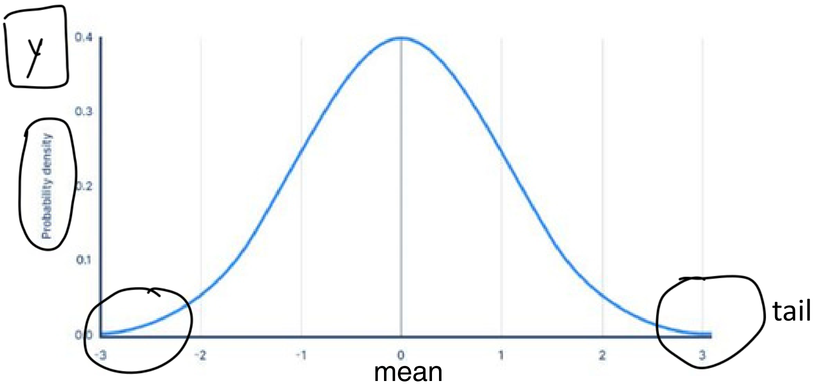
https://www.youtube.com/live/ADqYqSdtyW8?si=ddgBbb06QoqM9B-M

# Normal Distribution

## 1.What is normal distribution?

Normal distribution, also known as Gaussian distribution, is a probability distribution that is commonly used in statistical analysis. It is a continuous probability distribution that is symmetrical around the mean, with a bell-shaped curve.




-> Tail
-> Asymptotic in nature
-> Lots of points near the mean and very few far away

The normal distribution is characterized by two parameters: the mean (μ) and the standard deviation (σ). The mean represents the centre of the distribution, while the standard deviation represents the spread of the distribution.

Denoted as:

$$X \sim N(\mu, \sigma)$$

$\mu \rightarrow$ mean

$\sigma \rightarrow$ std

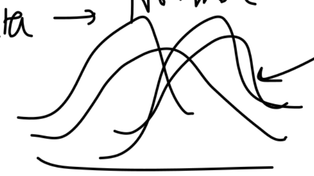
**Why is it so important?**

Commonality in Nature: Many natural phenomena follow a normal distribution, such as the heights of people, the weights of objects, the IQ scores of a population, and many more. Thus, the normal distribution provides a convenient way to model and analyse such data.

### PDF Equation of Normal Distribution

$$y = f(x) = \boxed{\frac{1}{\sigma\sqrt{2\pi}}}\,\boxed{e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}} \quad \boxed{\mu, \sigma}$$

data $\rightarrow$ Normal $\rightarrow$ $\boxed{\mu, \sigma}$



$\sigma\sqrt{2\pi}$

### Parameters in Normal Distribution

intr

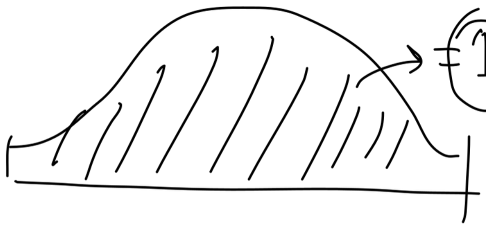
https://samp-suman-normal-dist-visualize-app-lkntug.streamlit.app/

### Equation in detail:

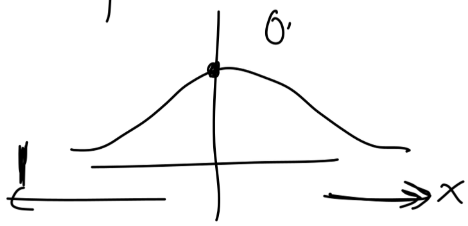
$$y = e^{-\frac{(x-\mu)^2}{2\sigma^2}} = \boxed{e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}}$$

$\sigma\sqrt{2\pi}$

$\sigma\sqrt{2\pi}$



$= 1$




$y = e^{-x^2}$

$y = \frac{1}{e^{x^2}}$

# Standard Normal Variate(Z) → standard normal distribution

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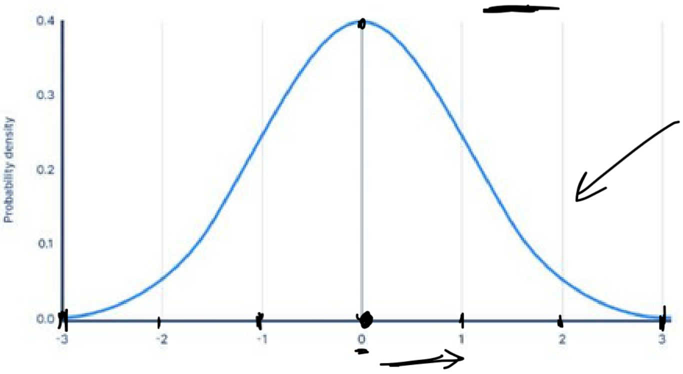
$X \sim N(\mu, \sigma)$ $\mu = 0$, $\sigma = 1$

$Z \sim N(0, 1)$

- What is Standard Normal Variate

A Standard Normal Variate(Z) is a standardized form of the normal distribution with mean = 0 and standard deviation = 1.

(Z)




Standardizing a normal distribution allows us to compare different distributions with each other, and to calculate probabilities using standardized tables or software.

Equation:

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2} = \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}x^2}$$

$X \sim N(5, 2.5)$

$\sigma = 1$, $\mu = 0$

- How to transform a normal distribution to Standard Normal Variate

$X \sim N(29, 14)$ → $N(0,1)$

Refer Python code

-2.5 0 2.5 5 7.5 10 12.5

-3 -2 -1 0 1 2 3

Kya Fayda Standardize karne ka?

Suppose the heights of adult males in a certain population follow a normal distribution with a mean of 68 inches and a standard deviation of 3 inches. What is the probability that a randomly selected adult male from this population is taller than 72 inches?

Age: 27, 25, 61, 73 → $\mu$, $\sigma$

$$Z = \frac{27 - \mu}{\sigma}$$

$X \sim N(68, 3)$ → (Z)

$$Z = \frac{72 - 68}{3} = \frac{4}{3}$$

55 62 65 68 71 74 77

0 1.33

1.33 → 0.90824 → 90.8

1.3, 0.03

$100 - 90.8 = 9.02$

- What are Z-tables

A z-table tells you the area underneath a normal distribution curve, to the left of the z-score

https://www.ztable.net/

For a Normal Distribution X~(u,std) what percent of population lie between mean and 1 standard deviation, 2 std and 3 std?

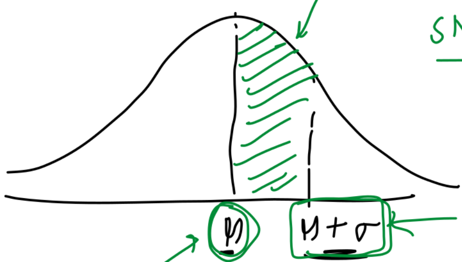
$X \sim N(\mu, \sigma)$ %

$\mu \rightarrow$

$Z = X - \mu$

standard deviation, 2 std and 3 std?

$X \sim N(\mu, \sigma)$ %



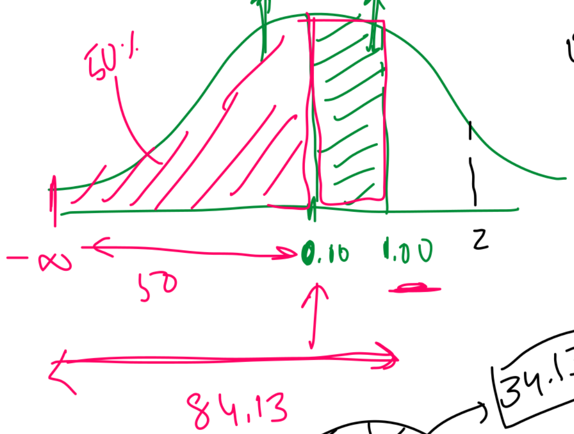

$\mu \rightarrow$

$Z = \frac{X-\mu}{\sigma}$

SNV(z) →

$z = \frac{\mu-\mu}{\sigma} = 0$

$Z = \frac{X-\mu}{\sigma} = \frac{\cancel{\mu}+\sigma-\cancel{\mu}}{\sigma} = \frac{\sigma}{\sigma} = 1$



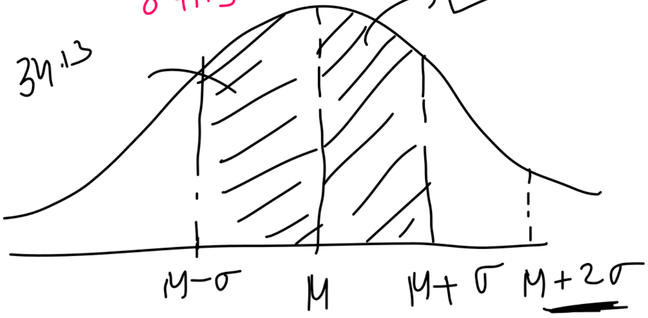

0.9772

97%

$84.13 - 50 = \boxed{34.13}$

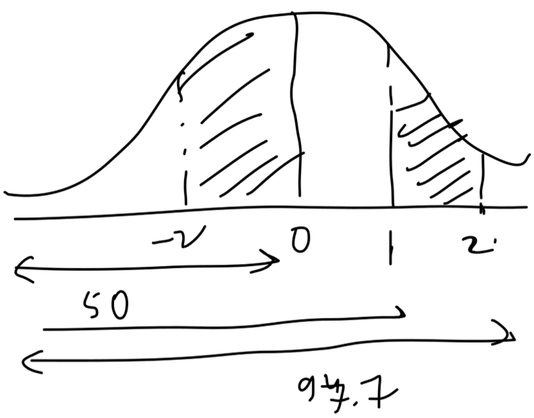
$-1\sigma$ $+1\sigma$

$\boxed{64.28}$ %




$z = \frac{\cancel{\mu}+2\sigma-\cancel{\mu}}{\sigma} = 2$

69.



# Properties of Normal Distribution

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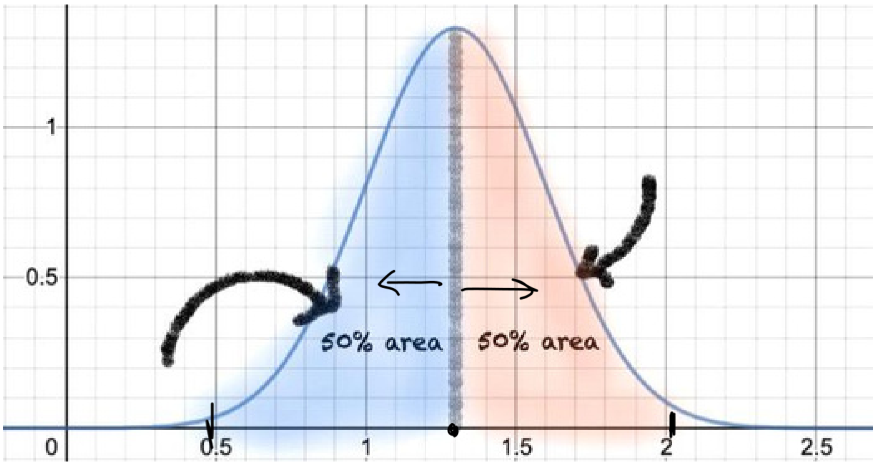
## 1. Symmetricity

The normal distribution is symmetric about its mean, which means that the probability of observing a value above the mean is the same as the probability of observing a value below the mean. The bell-shaped curve of the normal distribution reflects this symmetry.




## 2.Measures of Central Tendencies are equal

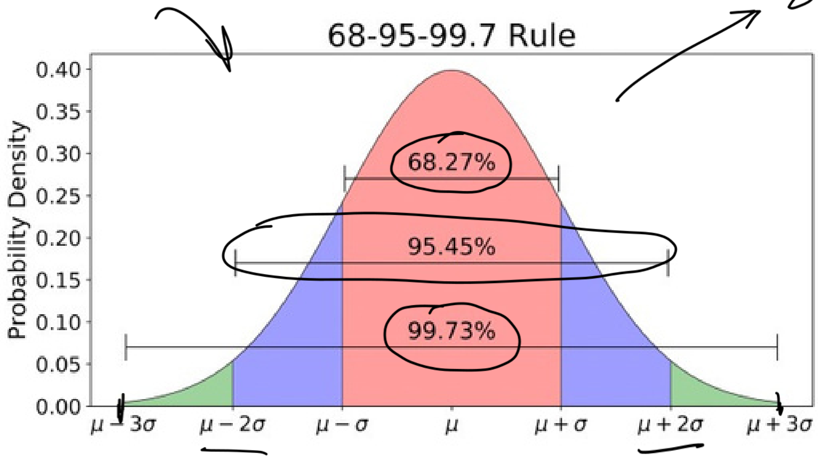
→ mean → median → mode

## 3.Empirical Rule

The normal distribution has a well-known empirical rule, also called the 68-95-99.7 rule, which states that approximately 68% of the data falls within one standard deviation of the mean, about 95% of the data falls within two standard deviations of the mean, and about 99.7% of the data falls within three standard deviations of the mean.

Standard normal variate

Z-table

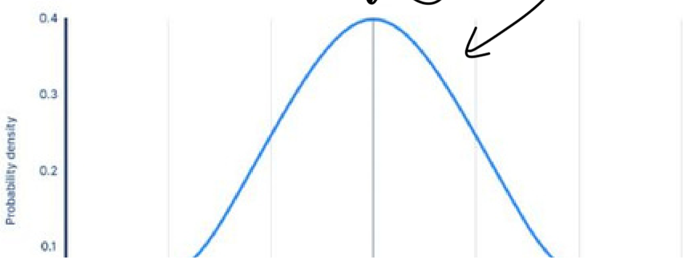


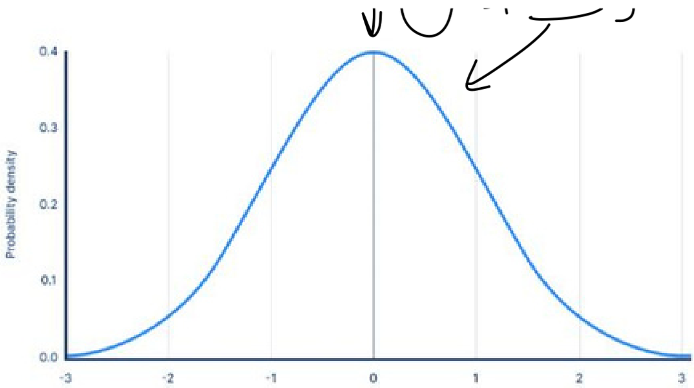


5σ

## 4.The area under the curve

1 → $\boxed{\text{PDF}}$




0.4
0.3
0.2
0.1
0.0
Probability density
-3
-2
-1
0
1
2
3

# Skewness

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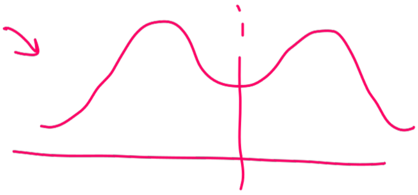


symmetric

- What is skewness?

A normal distribution is a bell-shaped, symmetrical distribution with a specific mathematical formula that describes how the data is spread out. Skewness indicates that the data is not symmetrical, which means it is not normally distributed.

Skewness is a measure of the asymmetry of a probability distribution. It is a statistical measure that describes the degree to which a dataset deviates from the normal distribution.

In a symmetrical distribution, the mean, median, and mode are all equal. In contrast, in a skewed distribution, the mean, median, and mode are not equal, and the distribution tends to have a longer tail on one side than the other.

Skewness can be positive, negative, or zero. A positive skewness means that the tail of the distribution is longer on the right side, while a negative skewness means that the tail is longer on the left side. A zero skewness indicates a perfectly symmetrical distribution.

moment

1 mom mean

2 mome — varia (2

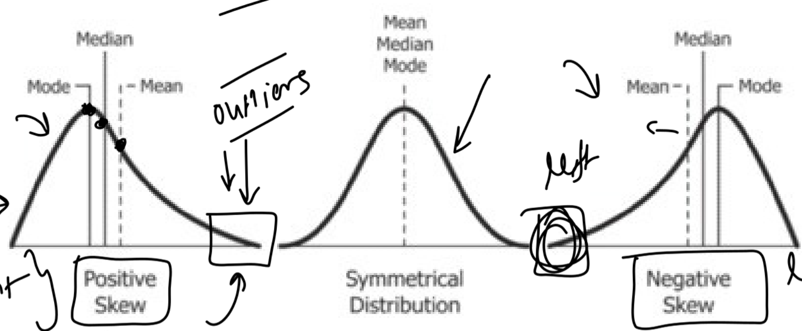
3 momnt → skew (3

4th → kurtosis

mode < median < mean

tail event

mode > median > mean




outliers

left

Right

left skewed

The greater the skew the greater the distance between mode, median and ~~mode~~ mean

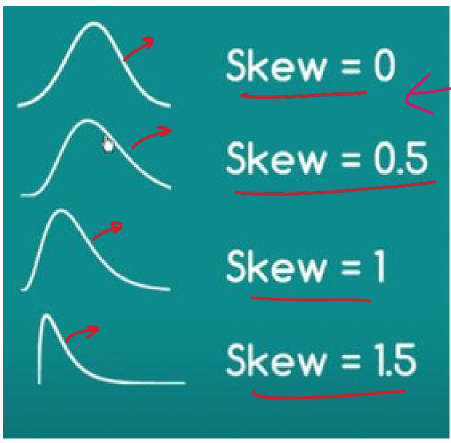
- How skewness is calculated?

moment

$$\frac{n}{(n-1)(n-2)}\sum\left(\frac{(x-\bar{x})}{s}\right)^3$$

sample skew

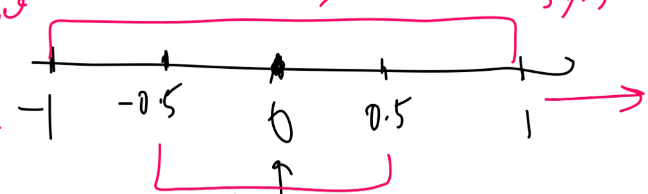
- Python Example
- Interpretation




highly skewe

0.38

moderately sym




almost sym

# CDF of Normal Distribution

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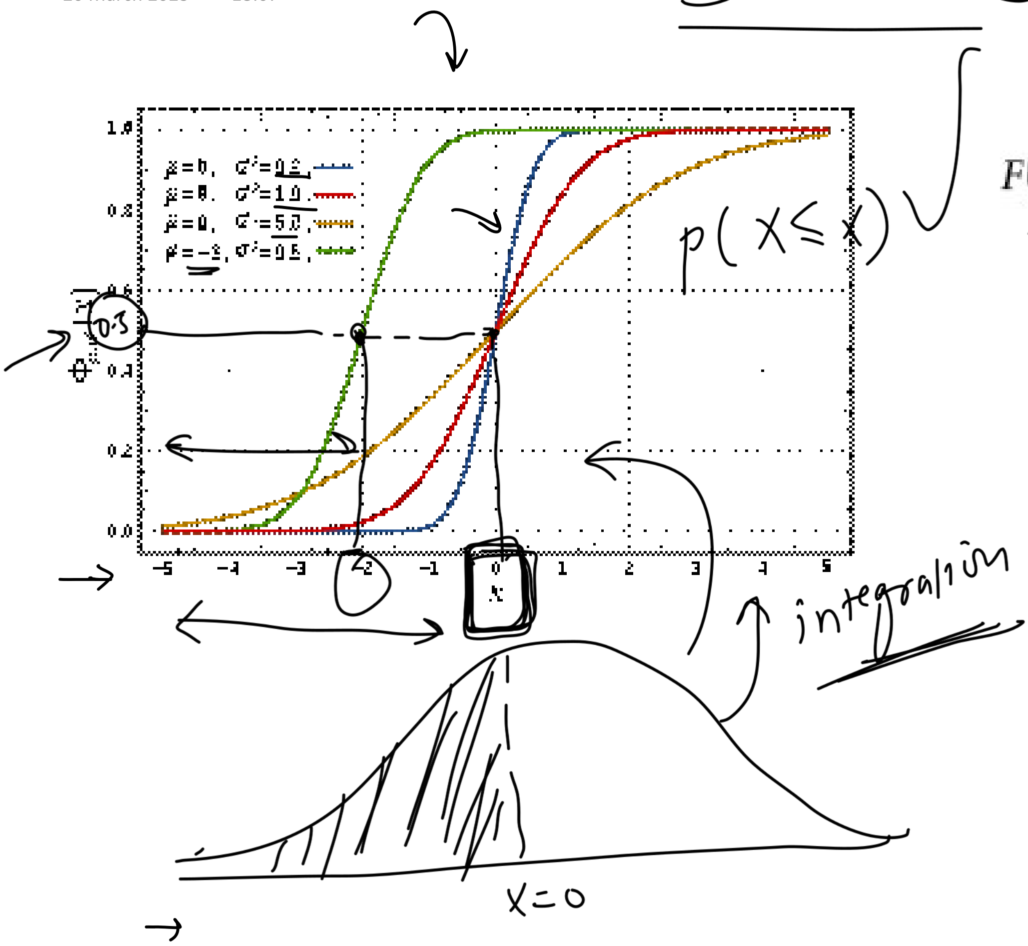


$$F(x) = p(X \le x) = \int_{-\infty}^{x} f(t)\,dt$$

$$= \frac{1}{\sigma\sqrt{2\pi}} \int_{-\infty}^{x} e^{-\frac{(t-\mu)^2}{2\sigma^2}}\,dt.$$

# Use in Data Science

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- Outlier detection
- Assumptions on data for ML algorithms -> Linear Regression and GMM
- Hypothesis Testing
- Central Limit Theorem